

Why is climate change an economic problem? (II)

— micro refresher + public goods

ECON-4210

Economics of Climate Change

Class 5

Prof. Casey Wichman

Today's class

1. Questions?
2. Attendance
3. Logistics: Ghost emissions // Problem Set 1 // Exam 1
4. **Climate change as an economic problem**
 - Review of key microeconomic concepts
 - Public goods and the commons
5. For next class...



Ghost Emissions

You run a business in one of nine boroughs in a city. You want to be profitable, but economic activity emits ghosts, which cause slime damage to everyone in the city, and containing them is costly. Each borough's mayor, rotating among the business owners, can set policies to contain ghosts and generate benefits (fewer ghosts released, less slime damage), but has to balance those benefits against the cost of containing them.

ECON 4210 • starts week 3

Set-up

Nine boroughs

One city, nine teams of five or six.
You are assigned to one when you sign up.

One business each

You manage your own business.
A bakery, laundry, brewery, freight depot, or foundry.

One shared environment

Ghosts from every business go into the same air. “Slime damage” affects all nine boroughs (differentially).

Translating ghosts to emissions...

Ghost emissions

greenhouse gases

Slime damage

climate damages

Ghost concentration

the stock in the atmosphere

Containment

abatement

The Ether

global temperature

The Institute

your Monday briefing

Two decisions a week

Everyone, every week

Your business

- What quantity to produce and sell.
- How many of your ghosts you want to contain.
- Containing costs you money. But, nobody pays you for it.

One person, rotating

Your borough

- Each week one teammate is mayor and submits for the team.
- Later rounds add a budget, a tax or a cap, and a treaty vote.
- Anyone can save a draft or send the mayor a suggestion.

Each week...

Monday

Results, and a briefing

During the week

Make your 1-2 decision(s)

Sunday 11:59 PM

Everything closes

Signing up

caseyjwichman.com/econ4210/ghost-emissions

Sign up with your GT email. Class code: **[CHECK CANVAS]**

How it counts

10 points

of your participation grade, out of 20 total

Two free misses

the cap is 10 points across 12 scored weeks

1 point

per week you submit your business decision

Rank is not graded

winning the city earns you nothing but glory

This week is practice. Nothing counts. Log in, check stuff out, move the sliders, submit.

Logistics (ii)

Problem Set 1

- Due: Friday, Sep 18th.

Exam 1

- Monday, Sep 21st
- Covers classes 1–6.
- Pen and paper, in class.
- One page of hand-written notes.

— Week 3 THIS WEEK

Sep 7–Sep 9

Mon Sep 7 *Labor Day — no class*

Wed Sep 9 **05**

Why is climate change an economic problem? (II) — the commons and the limits of bargaining

READINGS

- K&O Chs. 2, 4 & 5, continued — Canvas
- [Further reading:] Michelle Nijhuis (2021). "The miracle of the commons." *aeon*. [LINK](#)
- [Further reading:] Eugene A. Rosa and Thomas Dietz (2012). "Human drivers of national greenhouse-gas emissions." *Nature Climate Change*. [PDF](#)

GHOST EMISSIONS

- R0 · Practice — the game opens. Nothing counts this week.

— Week 4

Sep 14–Sep 16

Mon Sep 14 **06** What are the costs and benefits of climate policy?

READINGS

- K&O Ch. 3 — Canvas
- Kenneth J. Arrow et al. (1996). "Is there a role for benefit-cost analysis in environmental, health, and safety regulation?" *Science*. [PDF](#)

GHOST EMISSIONS

- R1 · Ghosts add up

Wed Sep 16 **Exam 1 review + flex**

In class. Review for Exam 1 — come with questions.

— Week 5

Sep 21–Sep 23

Mon Sep 21 **Exam 1** **EXAM**

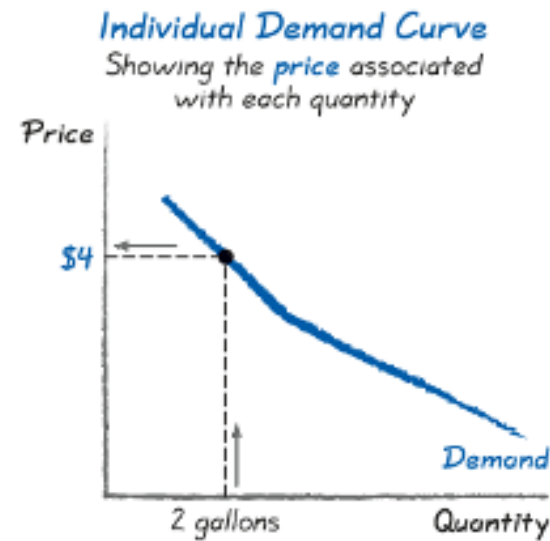
In class. Covers classes 1–6.

Key microeconomic concepts

- Marginal benefits and costs
- Aggregate demand and supply
- Equilibrium
- Consumer and producer surplus
- Tax incidence
- Externalities

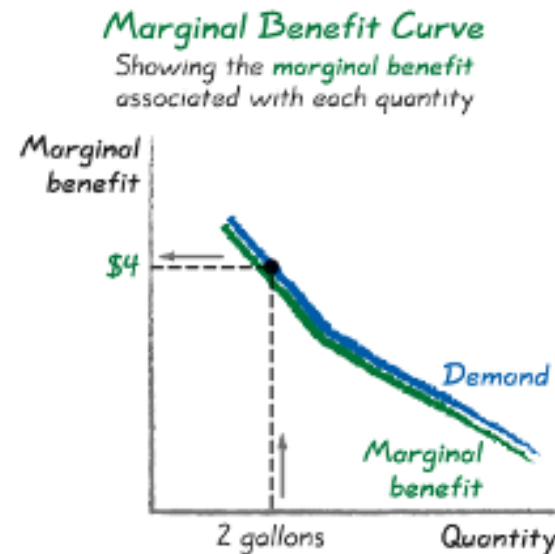
Marginal costs and benefits

- Marginal costs reflect the supply curve.
 - Supply curves are upward sloping because of diminishing marginal returns.
- Marginal benefits reflect the demand curve.
 - Demand curves are downward sloping because of diminishing marginal benefits.



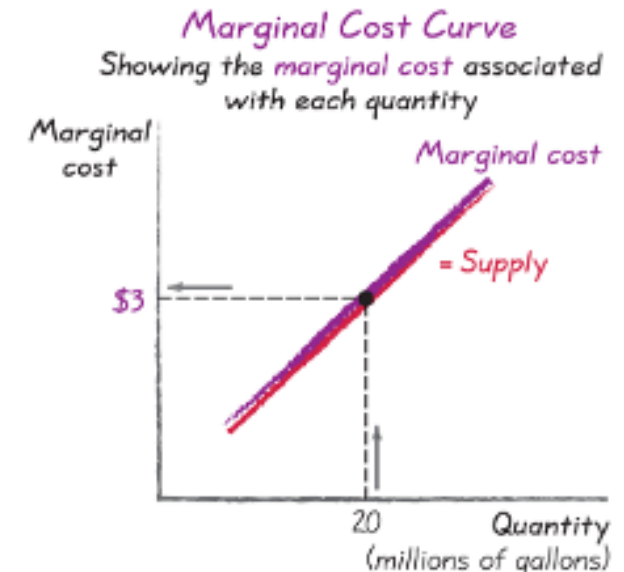
+

Price = Marginal benefit



+

Price = Marginal cost

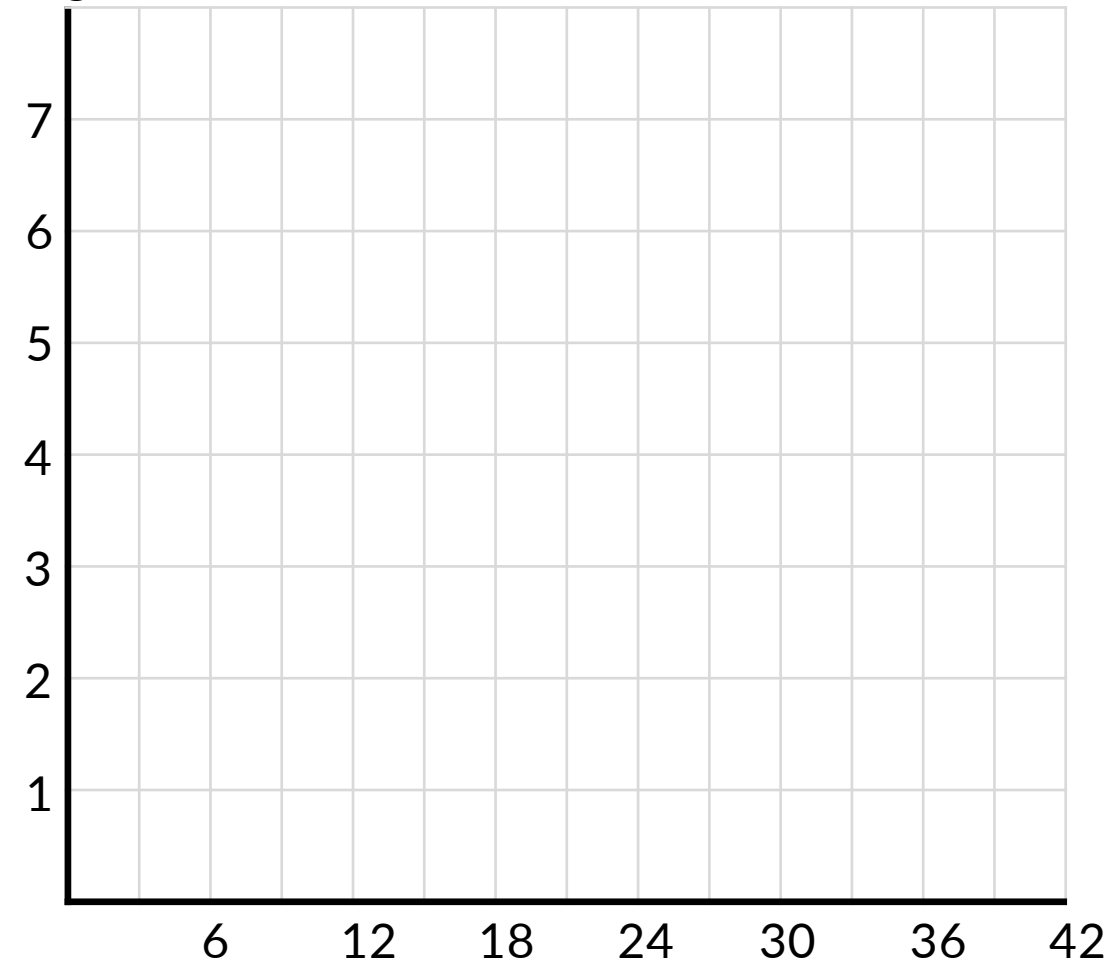


Aggregate demand (and supply)

- Individual demand curves summarize an individual consumer's willingness-to-pay for a good.
- Aggregate demand just adds up individual demand curves (horizontally at a given price)

Price	Quantity demanded of gasoline (gallons/week)			
	Person A	Person B	Person C	Aggregate
\$1	6	12	18	
\$2	5	10	15	
\$3	4	8	12	
\$4	3	6	9	
\$5	2	4	6	
\$6	1	2	3	
\$7	0	0	0	

Price
(\$/gallon)

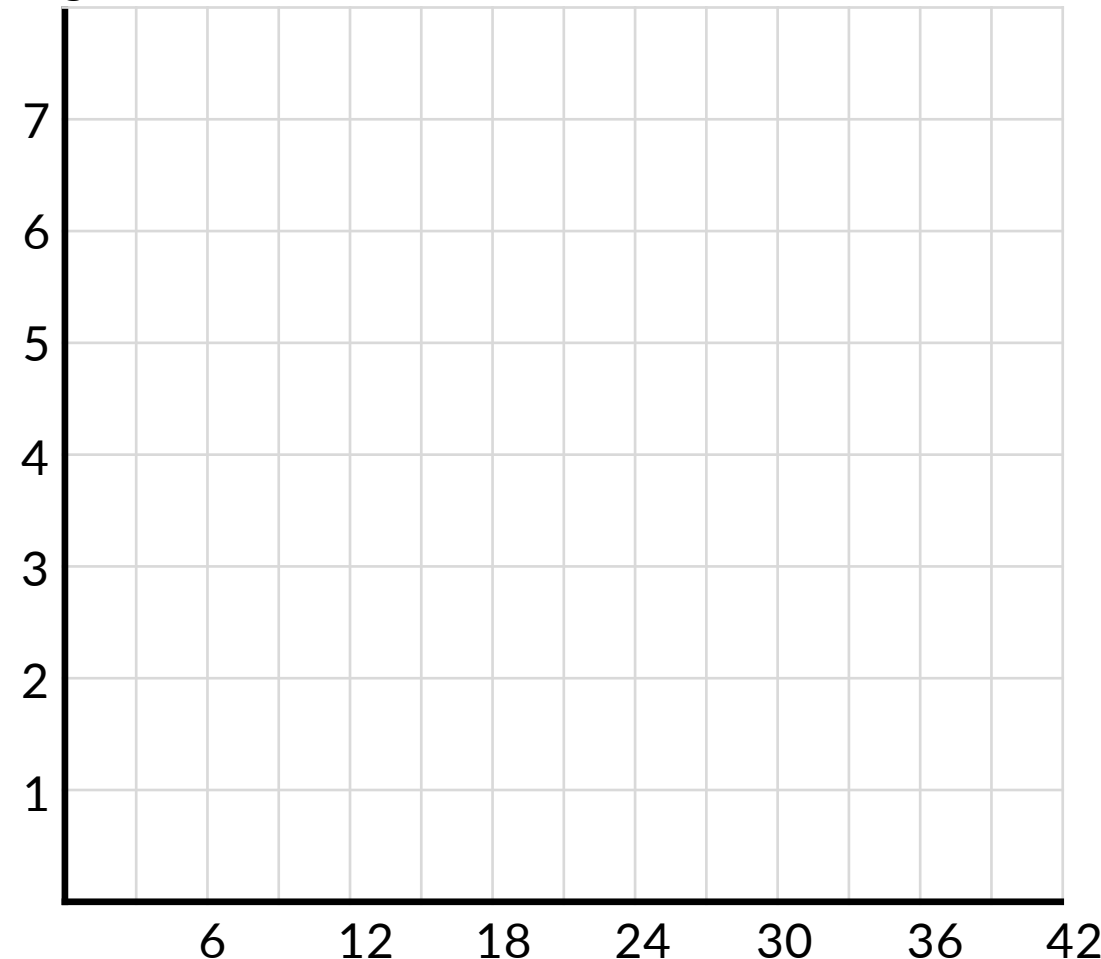


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Price	Quantity demanded of gasoline (gallons/week)			
	Person A	Person B	Person C	Aggregate
\$1	6	12	18	36
\$2	5	10	15	30
\$3	4	8	12	24
\$4	3	6	9	18
\$5	2	4	6	12
\$6	1	2	3	6
\$7	0	0	0	0

Price
(\$/gallon)



Aggregate demand (and supply)

- Individual demand curves summarize an individual consumer's willingness-to-pay for a good.
- Aggregate demand just adds up individual demand curves (horizontally at a given price)

Quantity demanded of gasoline (gallons/week)				
Price	Person A	Person B	Person C	Aggregate
\$1	6	12	18	36
\$2	5	10	15	30
\$3	4	8	12	24
\$4	3	6	9	18
\$5	2	4	6	12
\$6	1	2	3	6
\$7	0	0	0	0



Aggregate demand (and supply)

- Individual demand curves summarize an individual consumer's willingness-to-pay for a good.
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\$4	3	6	9	18
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\$6	1	2	3	6
\$7	0	0	0	0



Aggregate demand (and supply)

- We can do this with algebra, too...
- Let's define individual demand functions:
 - $MB_A(q): p = 7 - q$
 - $MB_B(q): p = 7 - 0.5q$
 - $MB_C(q): p = 7 - q/3$
- Find aggregate demand by **horizontally** summing individual demand curves: $Q = q_A + q_B + q_C$
 - Solve each MB curve for q
 - $q_A = 7 - p$
 - $q_B = 14 - 2p$
 - $q_C = 21 - 3p$
 - $\Rightarrow Q = [7 - p] + [14 - 2p] + [21 - 3p]$
 - $\Rightarrow Q = 42 - 6p$
 - (Solve for p): $p = 7 - Q/6$



Equilibrium

- Aggregate demand (or MB): $p = 7 - Q/6$
- Supply (or, marginal costs): $MC(q): p = 1 + q/12$



Equilibrium

- Aggregate demand (or MB): $p = 7 - Q/6$
- Supply (or, marginal costs): $MC(q): p = 1 + q/12$

- Equilibrium price / quantity?

- Set supply = demand (or MB=MC)

$$7 - q/6 = 1 + q/12$$

$$\Rightarrow 6 = (3/12)q \text{ or } q^* = 24$$

- Plug q^* into MC or MB

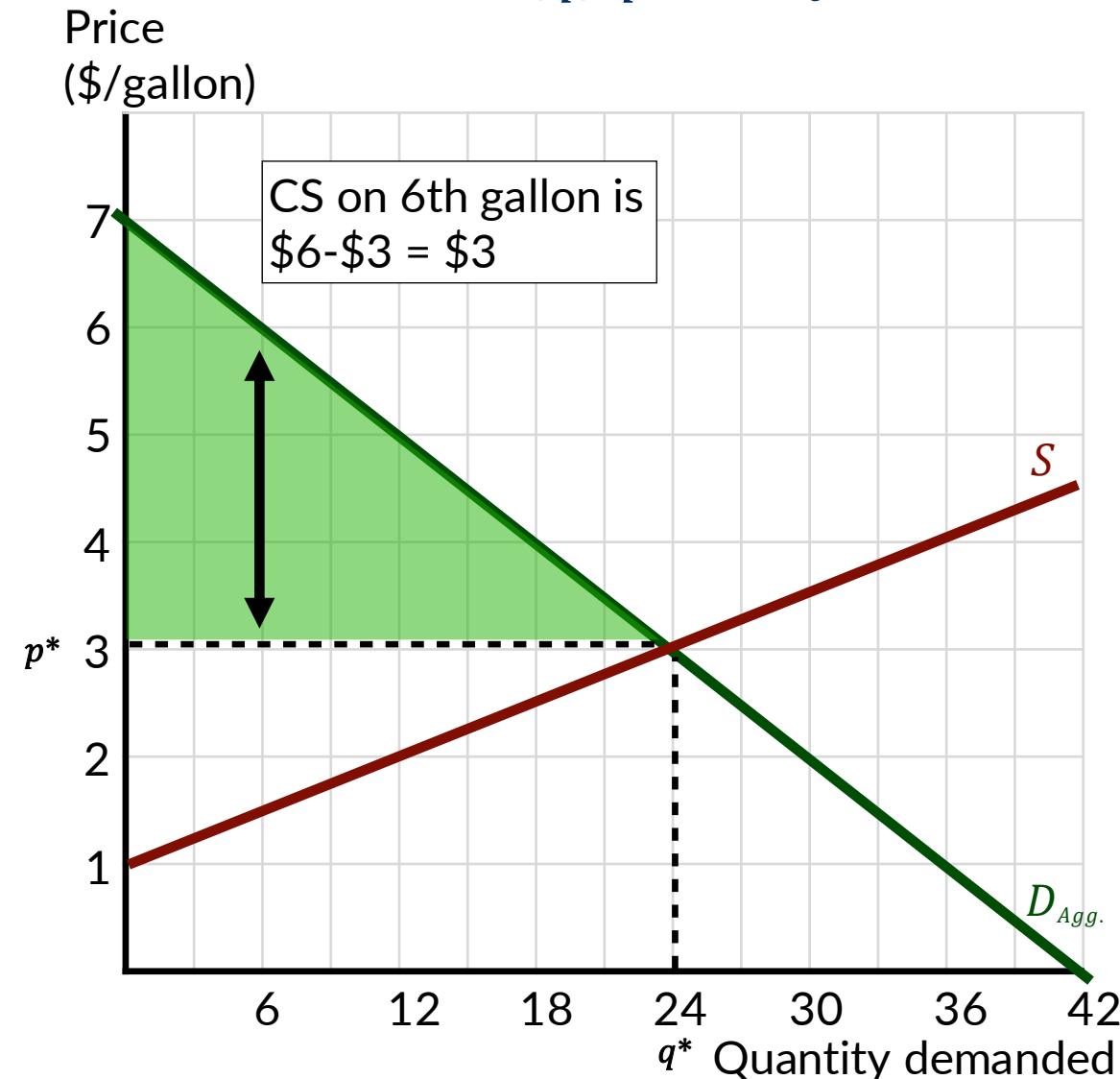
$$p^* = 7 - 24/6 = \$3/\text{gallon}$$



Consumer/producer surplus

- $MB(q): p = 7 - Q/6$
- $MC(q): p = 1 + Q/12$

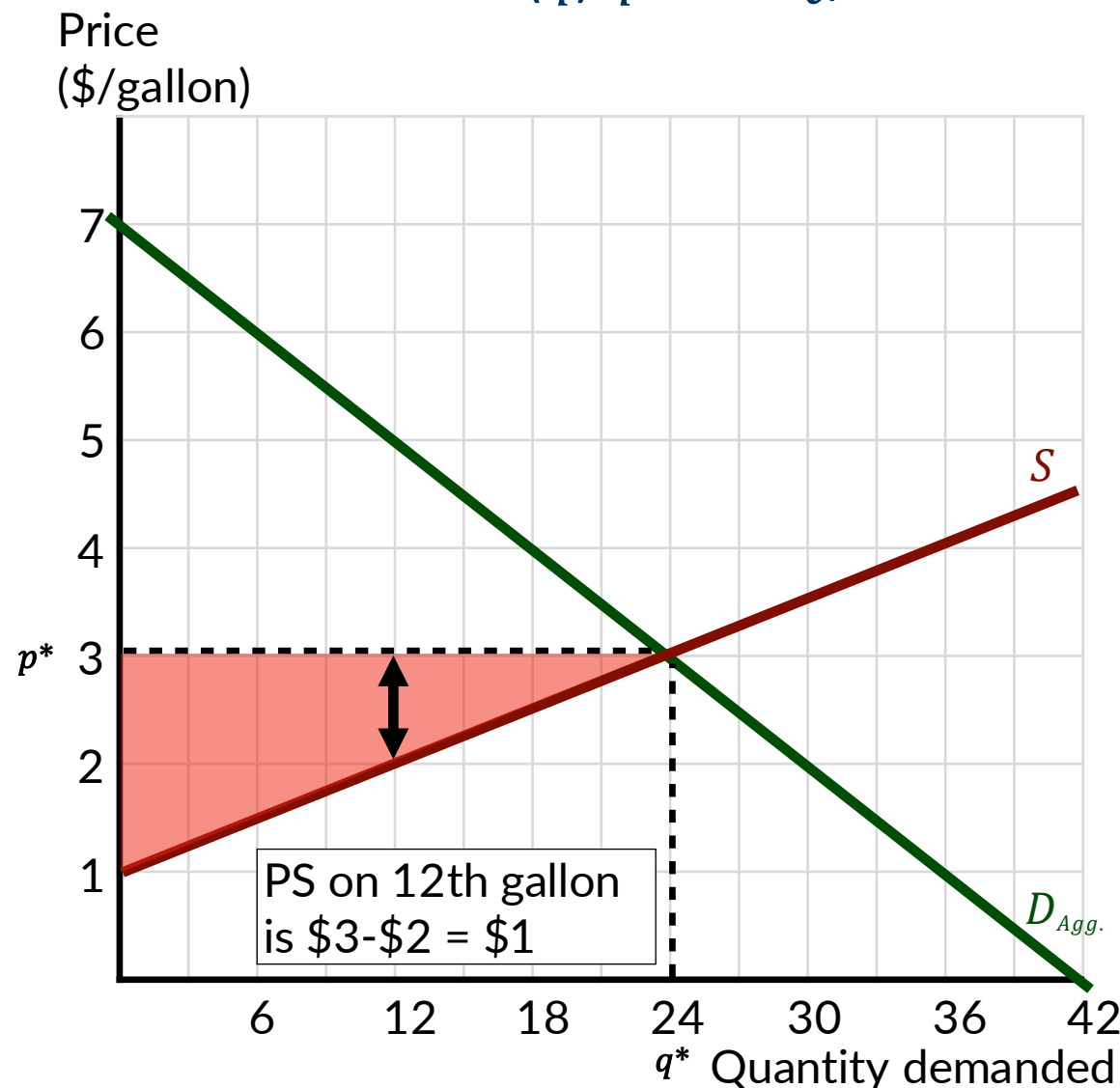
- The difference between marginal benefits (or costs) is called “economic surplus”
- Consumers only want to buy gasoline if $MB \geq p$.
- **Consumer surplus:**
 - ▶ The maximum consumers are willing to pay is given by their demand curve.
 - ▶ But, they only have to pay p^* on each unit.
 - ▶ So, each unit purchased generates economic surplus.
 - ▶ **CS** = Area below demand curve, but above price.
 - You can “integrate” or just find the area of a triangle:
 - $\frac{1}{2} \times \text{base} \times \text{height}$
 - ▶ $CS = \frac{1}{2} \times (24 - 0) \times (\$7 - \$3) = \48



Consumer/producer surplus

- The difference between marginal benefits (or costs) is called “economic surplus”
- Producers only want to sell gasoline if $MC \leq p$.
- **Producer surplus:**
 - ▶ The minimum producers are willing to sell gasoline for is given by the supply curve.
 - ▶ This reflects the marginal cost of producing each unit.
 - ▶ But, they receive p^* on every unit sold.
 - ▶ So, each unit sold generates economic surplus.
 - ▶ **PS** = Area above supply curve, but below price.
 - You can “integrate” or just find the area of a triangle
 - $\frac{1}{2} \times \text{base} \times \text{height}$
 - ▶ **PS** = $\frac{1}{2} \times (24 - 0) \times (\$3 - \$1) = \24

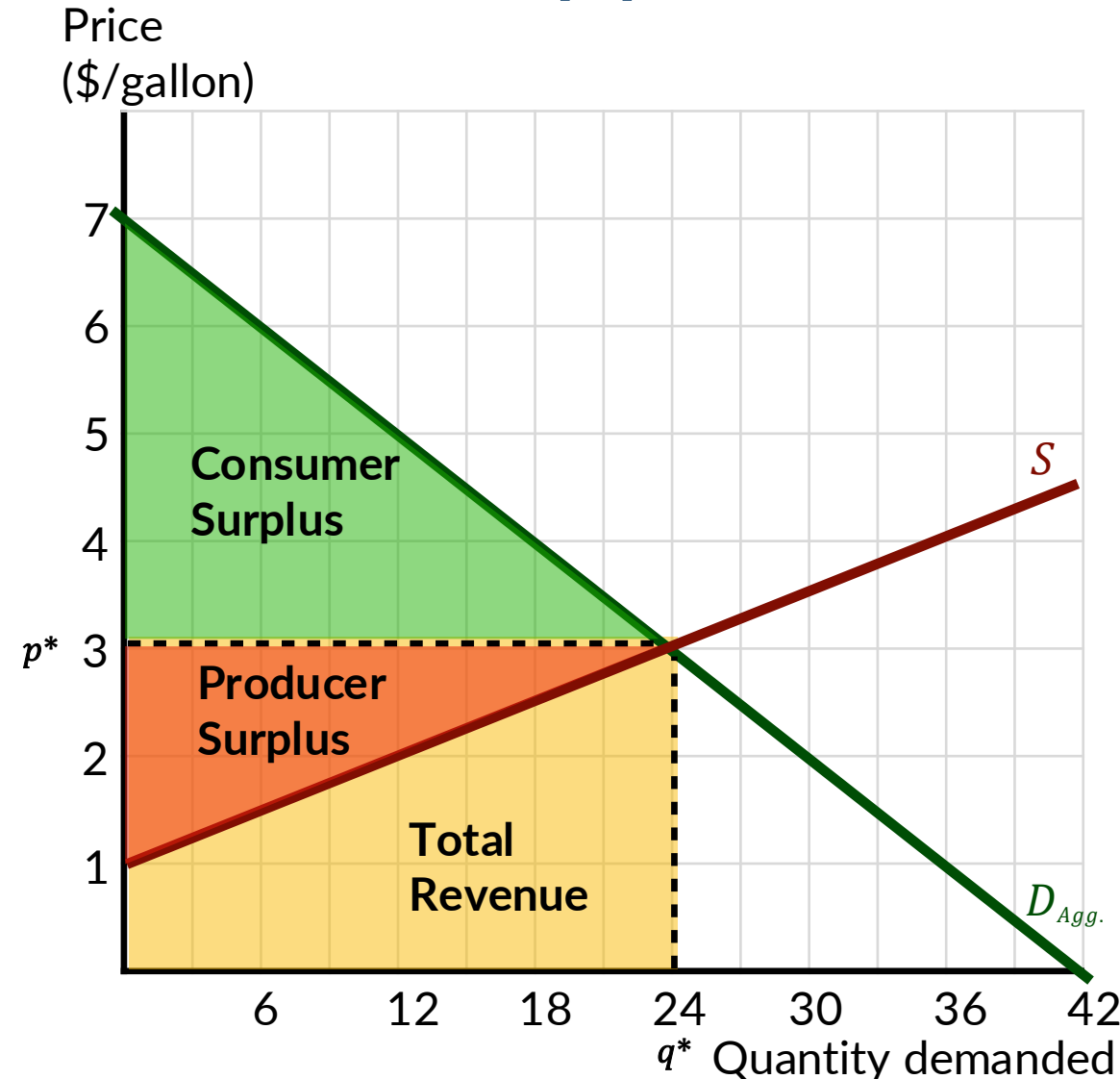
- $MB(q): p = 7 - Q/6$
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Consumer/producer surplus

- $MB(q): p = 7 - Q/6$
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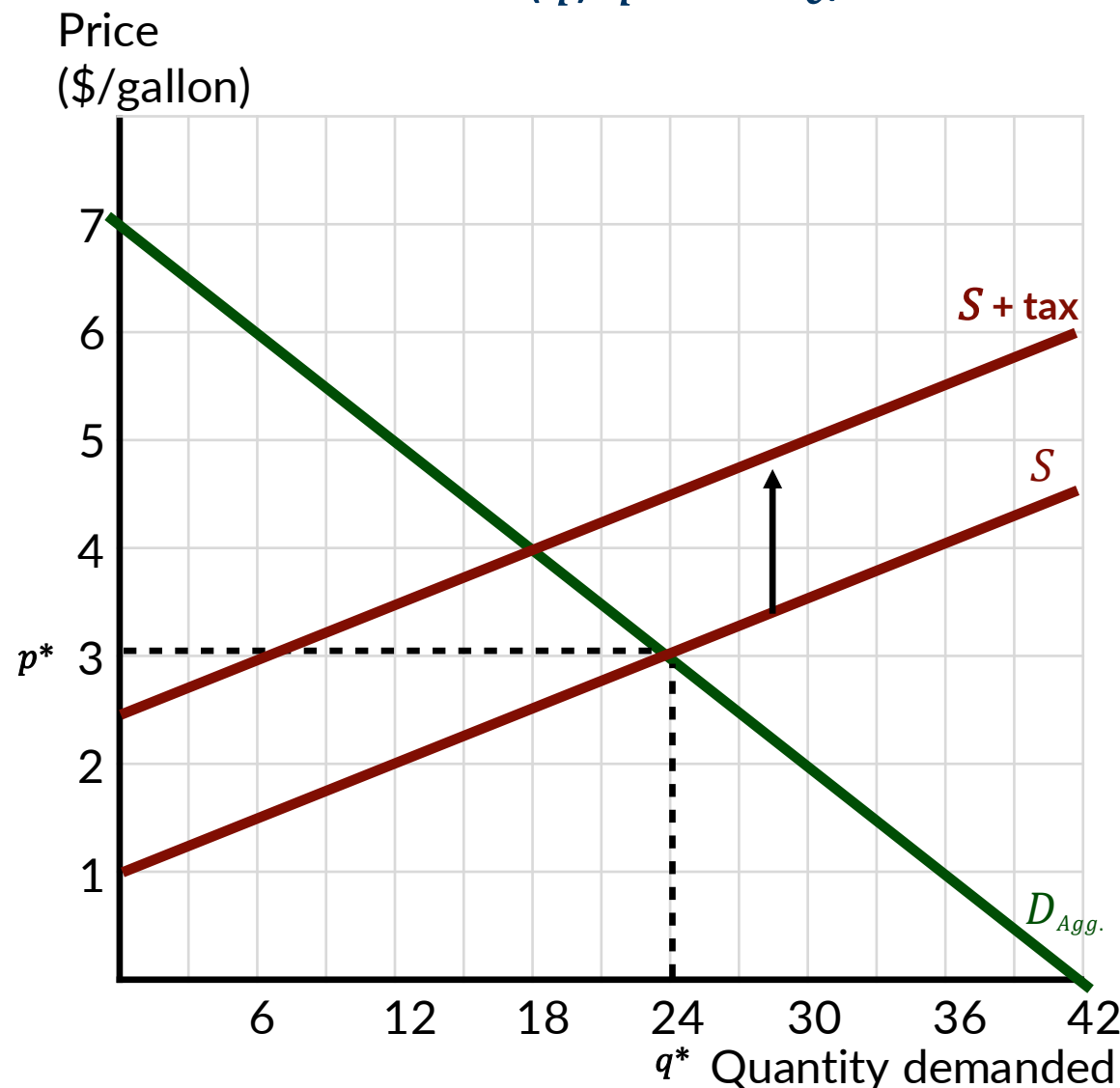
- *Triangles* tell us something about “welfare”
- **Consumer surplus** is the total economic benefit that is captured by consumers (essentially benefits minus costs).
- **Producer surplus** is the total economic benefit that is captured by producers (essentially benefits minus costs).
- *Rectangles* are also helpful.
 - **Total revenue** = $p^* \times q^* = \$3 \times 24 = \72
 - Additionally, this is total consumer expenditures.



Taxes

- Many policies/regulations affect market equilibrium.
- Taxes increase prices consumer pay and decrease prices suppliers receive.
- Imagine a \$1.50/gallon tax on gasoline producers.
 - Vertically shifts supply curve by \$1.50/gal

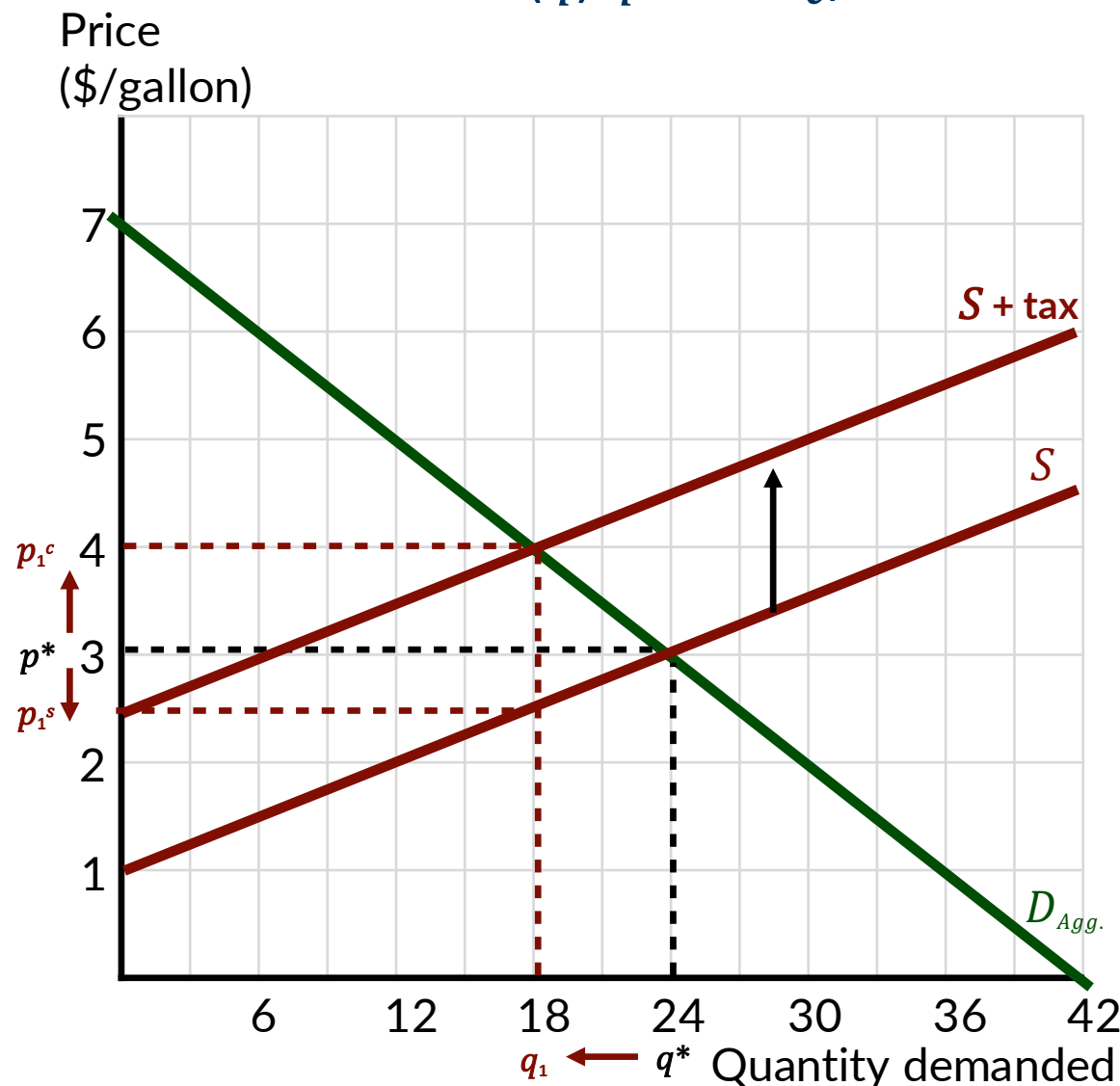
- $MB(q): p = 7 - Q/6$
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Taxes

- Many policies/regulations affect market equilibrium.
- Taxes increase prices consumer pay and decrease prices suppliers receive.
- Imagine a \$1.50/gallon tax on gasoline producers.
 - Vertically shifts supply curve by \$1.50/gal
 - New equilibrium:
 - $p_1^c = \$4$
 - $q_1 = 18$ gallons / week
 - Consumers pay \$4/gallon, BUT producers only receive \$2.50/gallon.
 - Where does the \$1.50/gallon go?

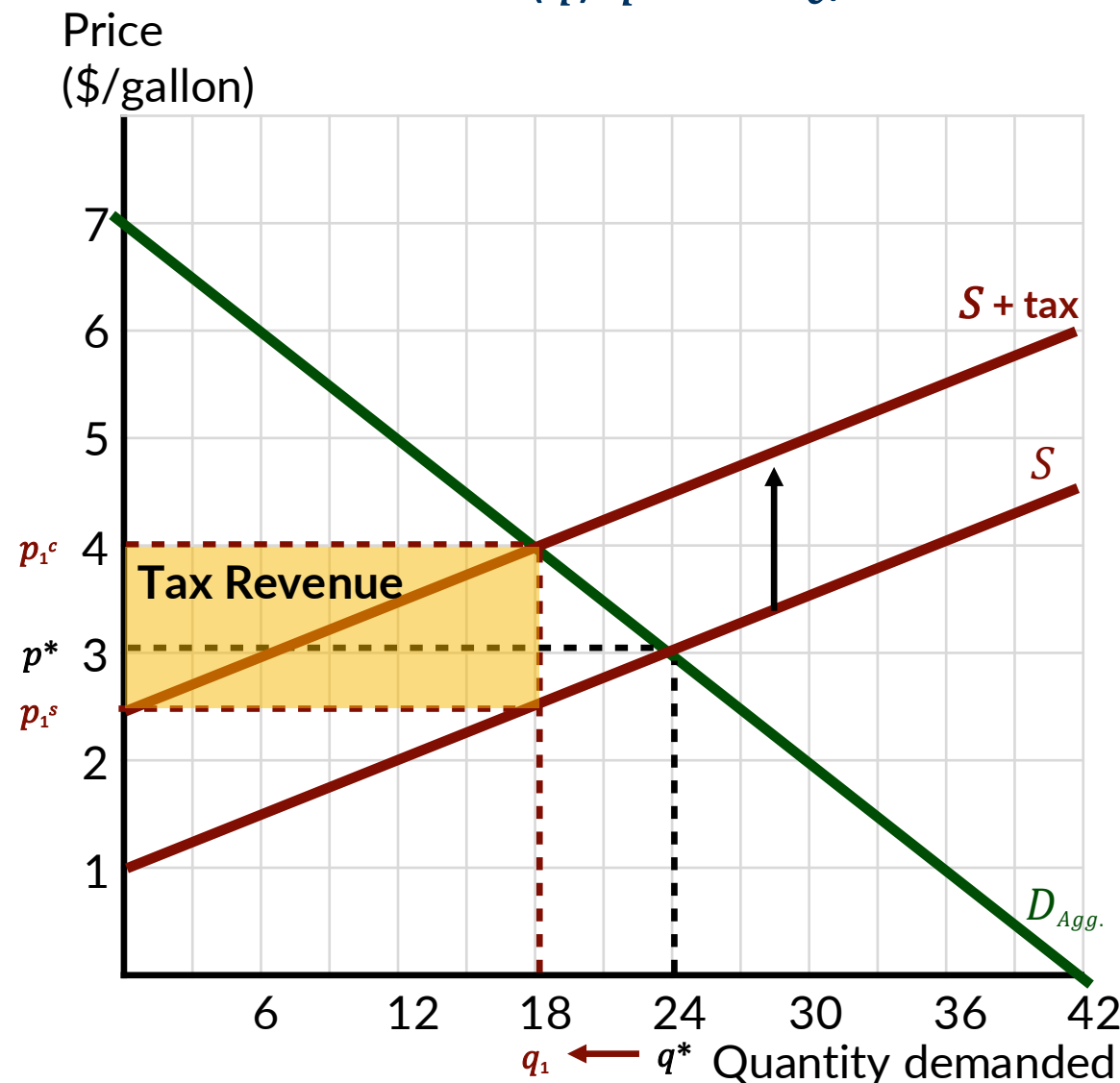
- $MB(q): p = 7 - Q/6$
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Taxes

- Consumers' **tax incidence** is the price they pay with the tax minus the original price (or, $\$4 - \$3 = \$1$).
- Producer's **tax incidence** is the original price minus the price they receive with the tax (or, $\$3 - \$2.50 = \$0.50$)
- \$1.50 tax on gasoline generates **government revenue**.
 - How much? Revenue = $\tau \times q_1 = \$1.50 \times 18 = \27
- So, consumers's **tax burden** is $\$1 \times 18 = \18
- Producers's **tax burden** is $\$0.50 \times 18 = \9
- Even though the **statutory burden** (i.e., who has to pay the tax) falls on producers, consumers pay more of the tax.
 - Who the tax is levied on doesn't matter!
 - Tax burdens depend on relative slope of the supply and demand curves
 - I.e., if consumers are relatively inelastic (demand curve is "steeper" than supply curve) $\Rightarrow$ greater tax burden on consumers.

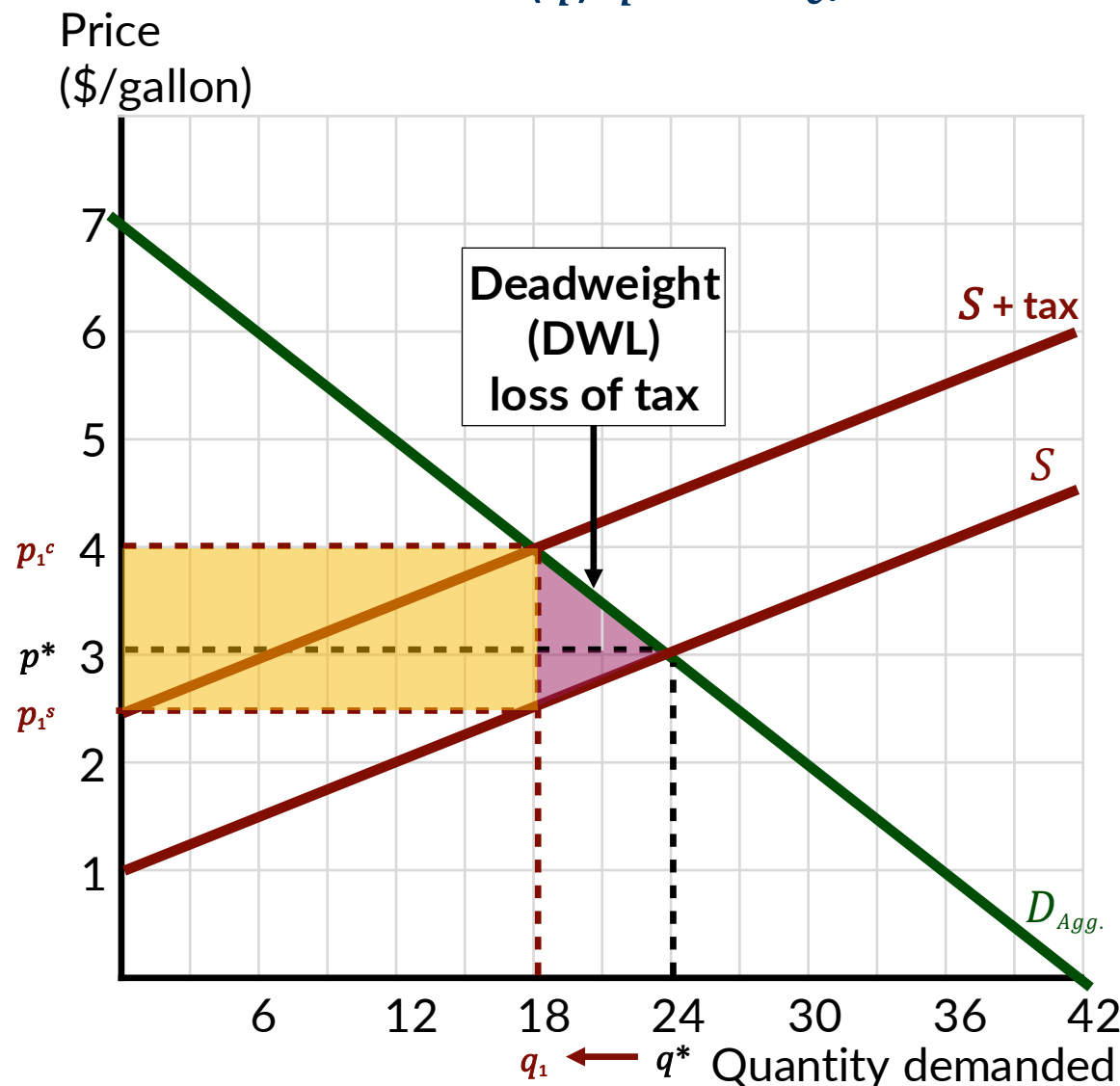
- $MB(q): p = 7 - Q/6$
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Taxes

- Anything that pulls a market away from equilibrium generates welfare costs.
- With the tax, we're consuming/producing less gasoline, which means we are "losing" benefits relative to the competitive
 - The **deadweight loss (DWL)** of the tax is the area of lost consumer + producer surplus that occurs from the tax.
 - **DWL of tax** = $\frac{1}{2} \times (24 - 18) \times (\$1.50) = \$4.50$
 - DWL of tax used to be consumer + producer surplus, but it was eliminated by the tax.
 - ***Importantly, externalities also generate DWL, which is represented by a different triangle.***

- $MB(q): p = 7 - Q/6$
- $MC(q): p = 1 + Q/12$



Externalities

- An **externality** results when the actions of one individual (or firm) have a direct, unintentional, and uncompensated effect on the well-being of other individuals or the profits of other firms.

- $MB(q): p = 7 - Q/6$
- $MC(q): p = 1 + Q/12$

Table 2. Summary of External Cost Estimates (2021\$)

External damages category	National average (per gallon)	National average (per mile)	Range across counties	Comparison with previous estimates
Carbon dioxide emissions	\$1.71	\$0.07	None	Well above previous estimates due to higher social cost of carbon (Rennert et al. 2022)
Local pollution	\$0.79	\$0.03	\$0.03–\$8.25 per gallon	Similar to previous estimates, e.g., Parry et al. (2007)
Congestion	\$0.27	\$0.01	\$0.00–\$0.12 per mile	Similar to previous national-level estimates, but we show substantial variation by county
Accidents	\$1.08	\$0.05	None	Estimate is from the literature; consistent with most other studies
Sum	\$3.85	\$0.16	\$2.83–\$13.44 per gallon \$0.12–\$0.56 per mile	Higher than most previous estimates, with wider range

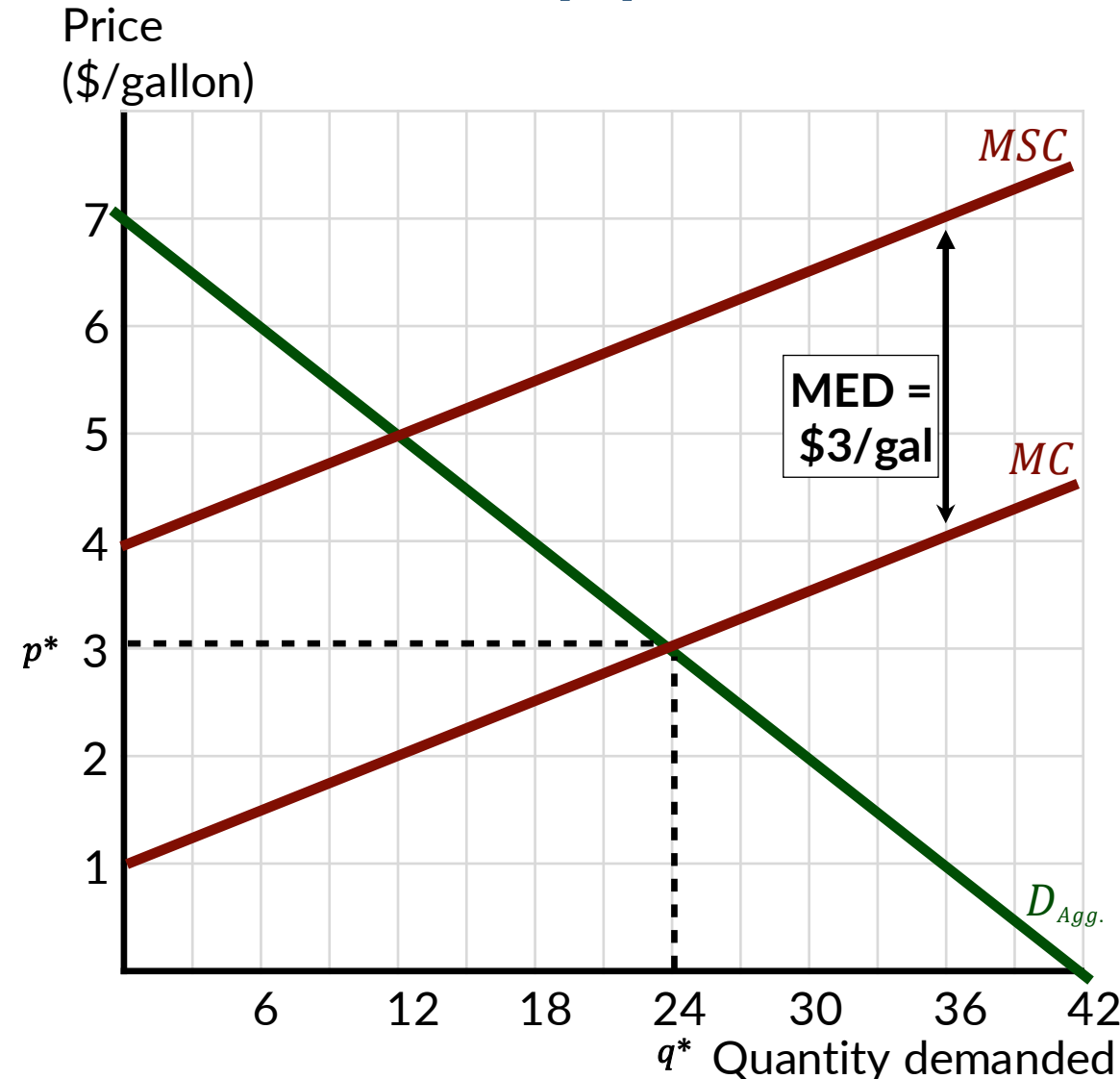
Notes: Average pollution and congestion damages are weighted by county-level population. GHG damages are uniform across counties. Accident externalities are assumed to be equal across counties because of a lack of data.



Externalities

- An **externality** results when the actions of one individual (or firm) have a direct, unintentional, and uncompensated effect on the well-being of other individuals or the profits of other firms.
- Assume that every gallon of gasoline consumed generates \$3 in **marginal external damages (MED)**.
- We model externalities by shifting the supply curve (vertically) by the level of the externality.
 - **Marginal Social Costs (MSC) = MC + MED**
 - The MSC curve isn't "real" because externalities are, by definition, not captured in the market.

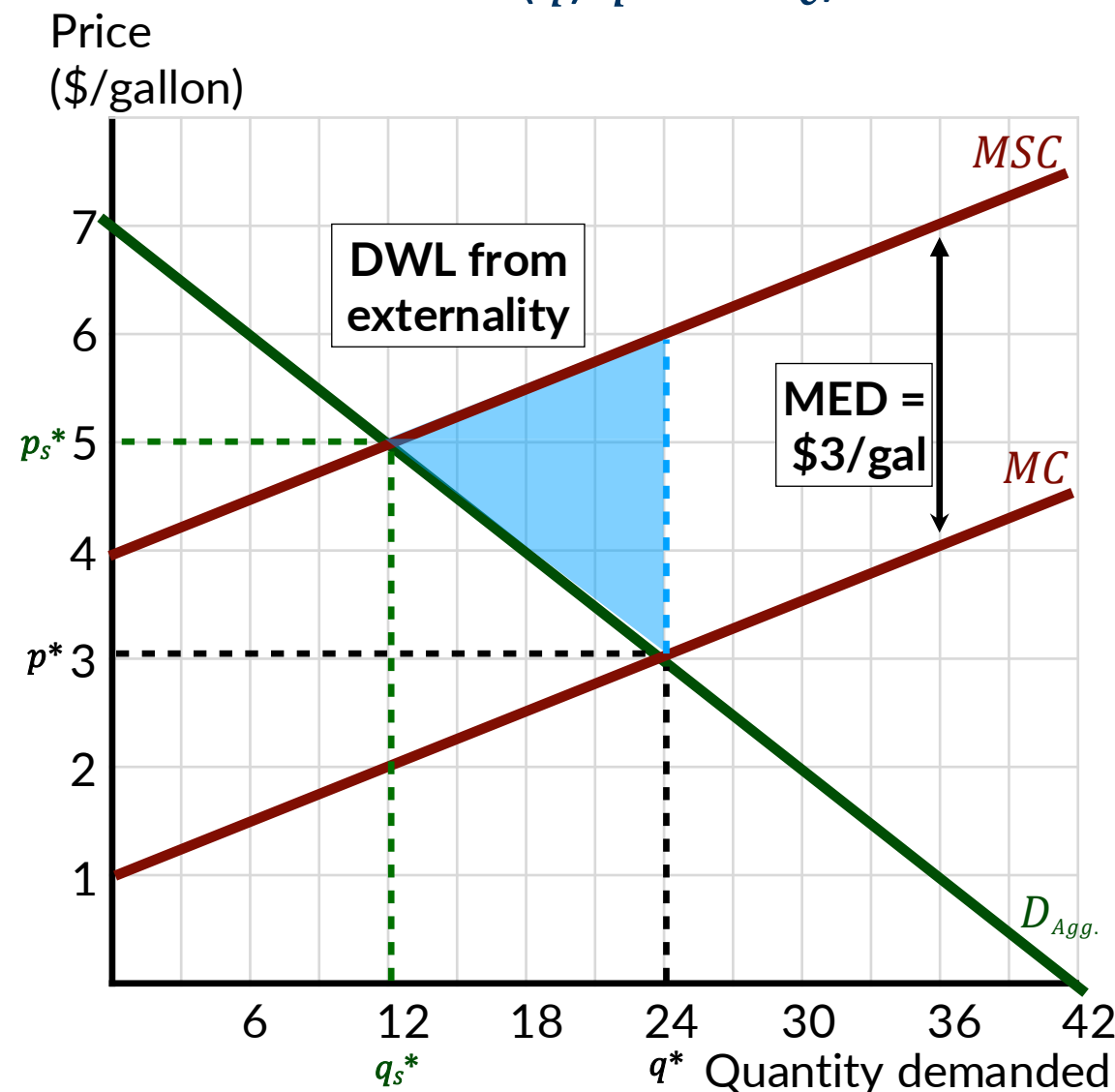
- $MB(q): p = 7 - Q/6$
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Externalities

- Externalities generate economic harm.
 - At $q^* = 24$, externalities generate \$3 worth of damages to actors *outside* of the market.
- If we “internalized” the external costs, we would arrive at a new **social equilibrium**.
 - Efficiency is now defined by $MB = MSC$.
 - New equilibrium price that internalizes social costs = \$5
- But, unaccounted for, externalities generate **deadweight loss**.
 - This is the total economic cost borne by actors outside of the market.
 - This is the area of above the demand curve that captures external damages.
- Much of environmental economics is focused on policies that get us to the social equilibrium cost-effectively.
 - We'll return to this later in the semester!

- $MB(q): p = 7 - Q/6$
- $MC(q): p = 1 + Q/12$



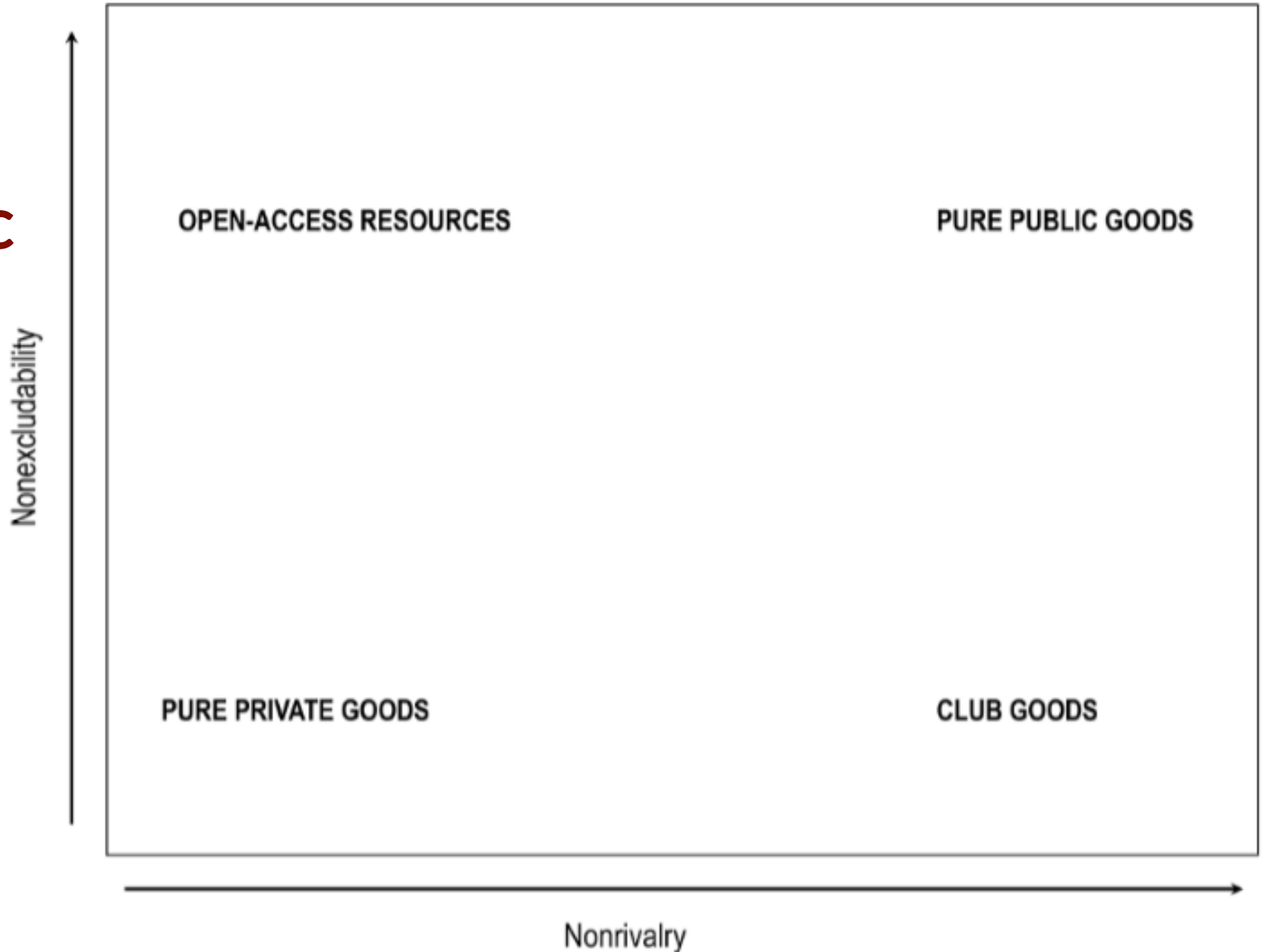
Public goods

What are they?

- **Public goods** are economic goods that are nonrival and nonexcludable.
- **Nonrival** means that my consumption of the good does not limit the amount or quality of the good that you consume.
- **Nonexcludable** means that you cannot be prevented from consuming the public good.
- We can also have **public “bads”** with the same properties
- GHG gas abatement that the US provides is a public good.
 - Our enjoyment of a slightly cooler planet doesn't prevent other countries from enjoying the benefits of a cooler planet.
 - We can't keep Mexico or France from enjoying the benefits of our GHG abatement (i.e., less climate change)

There is a
spectrum of
private/public
/club goods...

Can we fill
some in with
examples?



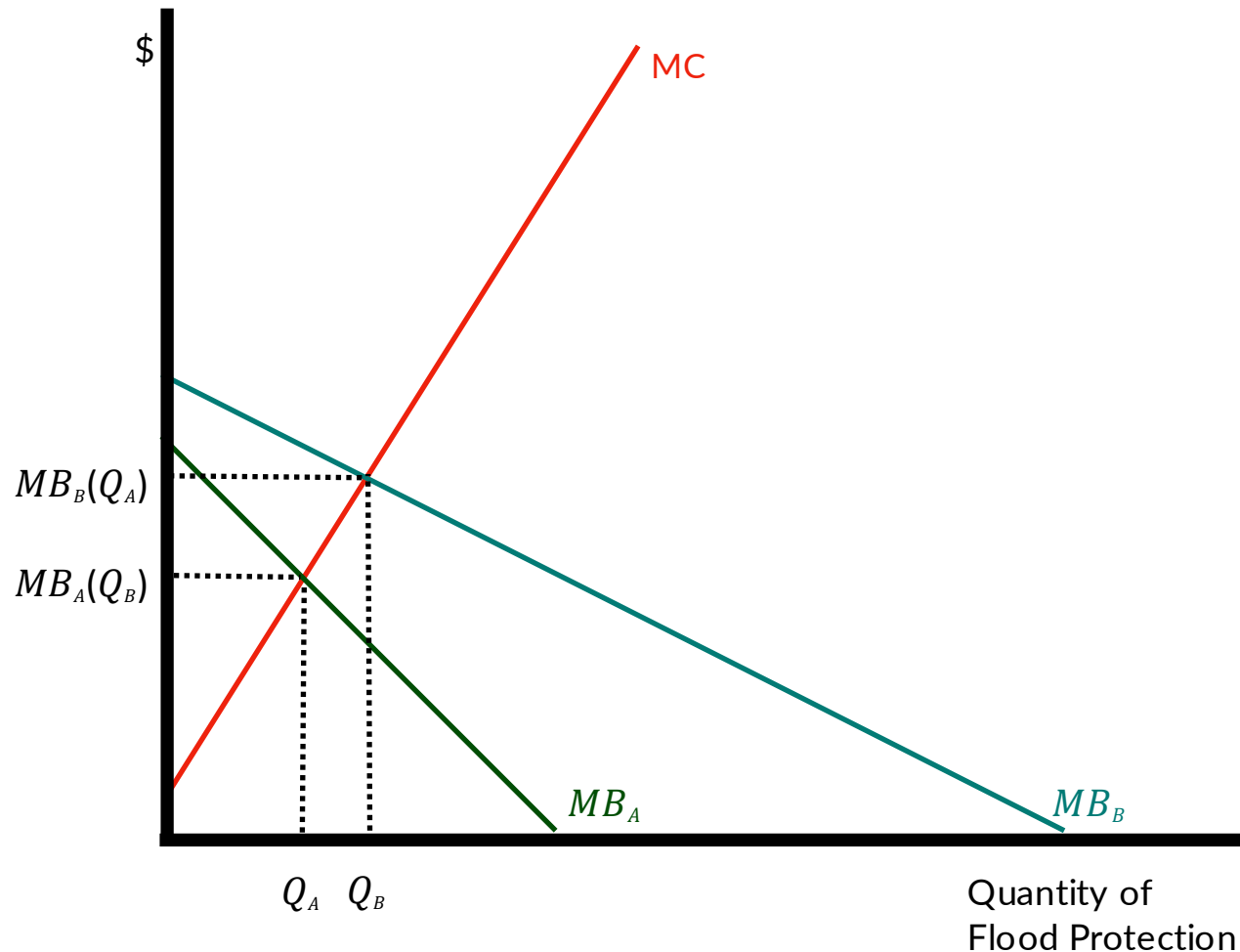
The problem with public goods

- **The free-rider problem.**

- ▶ A market failure that occurs because private incentives to free-ride on others' contributions results in under-provision of the public good.
 - ▶ Non-excludability means that I can consume the public good without having to pay for it.
 - ▶ Non-rivalry means that I my consumption of the public good is not diminished by others.
- If I think other countries are going to “pick up the slack” in delivering ambitious GHG emissions reductions, perhaps I can enjoy those benefits without investing in costly abatement at home.

Public goods

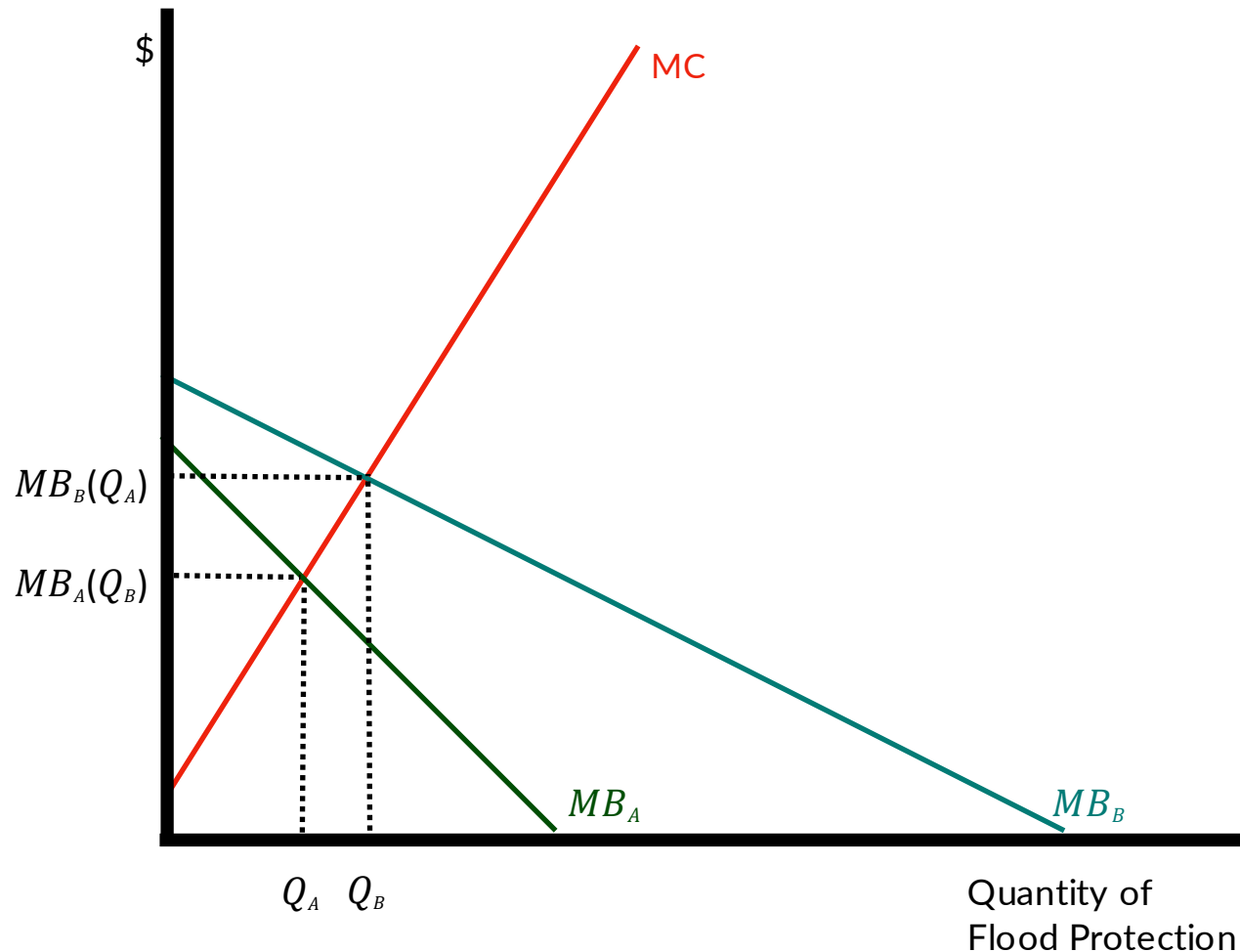
How do we model them?



- Flood protection (via dams/levees) offer flood protection from hurricanes.
- Consider two people (A & B) who receive different marginal benefits from investment in flood protection.
- A and B live in same town, but B's house is at lower elevation (higher expected flood risk) and has higher marginal benefits of flood protection
 - $MB_B > MB_A$
- Question to both consumers is:
 - *How much would you, individually, be willing to pay to provide flood protection?*

Public goods

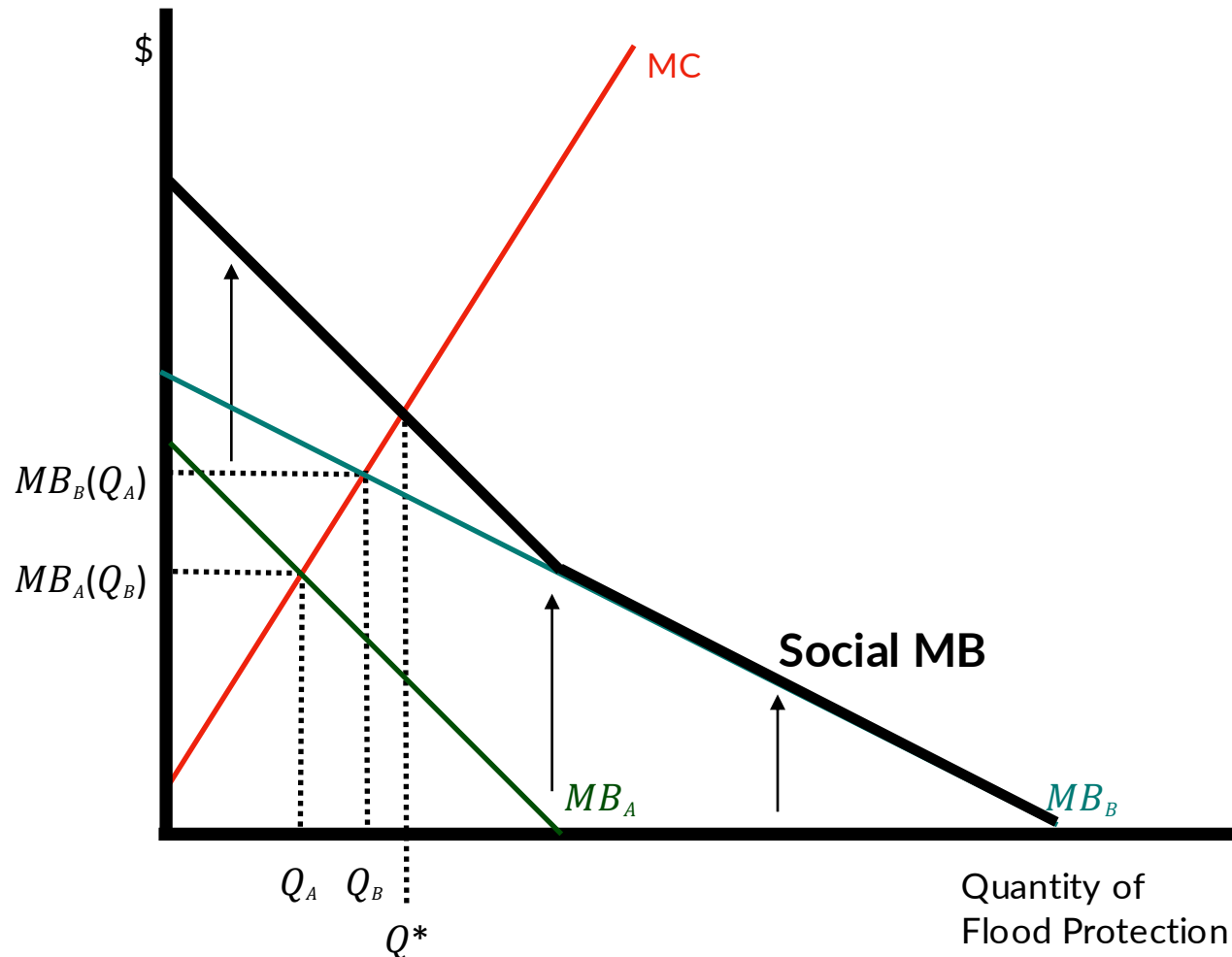
How do we model them?



- How much would you, individually, be willing to pay to provide flood protection?
- Left to own devices, A would only pay for protection up to point Q_A (i.e., where $MB_A = MC$)
- But, B would be willing to provide up to Q_B .
- Because flood protection is *nonrivalrous* & *nonexcludable*, the community would provide Q_B .
 - Maybe costs are split between A and B.
 - Maybe B pays for all of it, and A pays nothing.
- Is socially optimal level of provision obtained?

Public goods

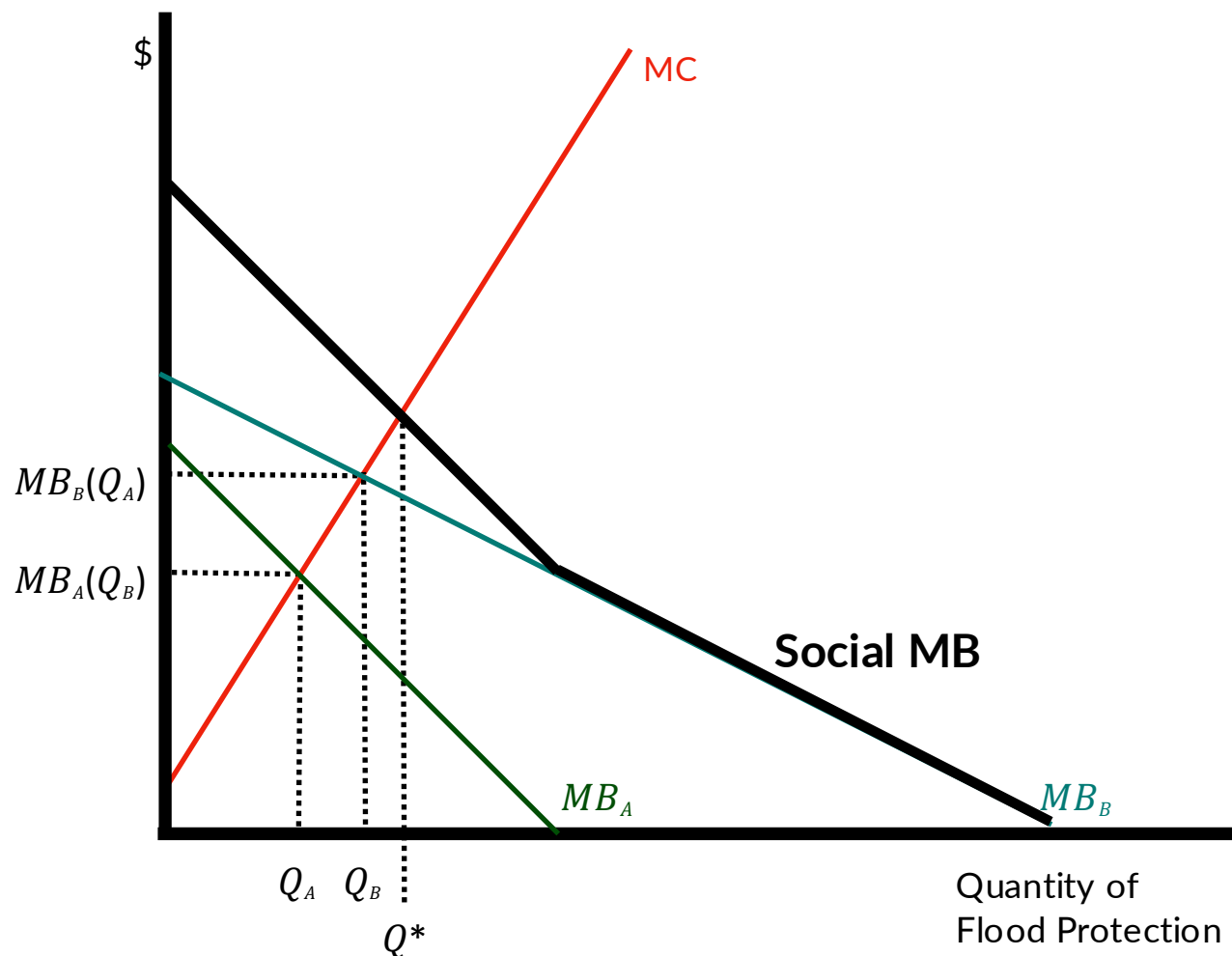
How do we model them?



- We need to add demand curves “vertically” to obtain social MB (or total willingness to pay for the public good).
- In the previous example, A had an incentive to free-ride on B’s provision, so the optimal amount of Q was underprovided.
 - This is suboptimal because we are not accounting for the benefits that accrue to both parties.
 - Economic surplus is not maximized.
- We see this because if we add up the marginal benefits curves vertically to define Social MB, we arrive at Q^* , where $Q^* > Q_B$.

Public goods

How do we model them?



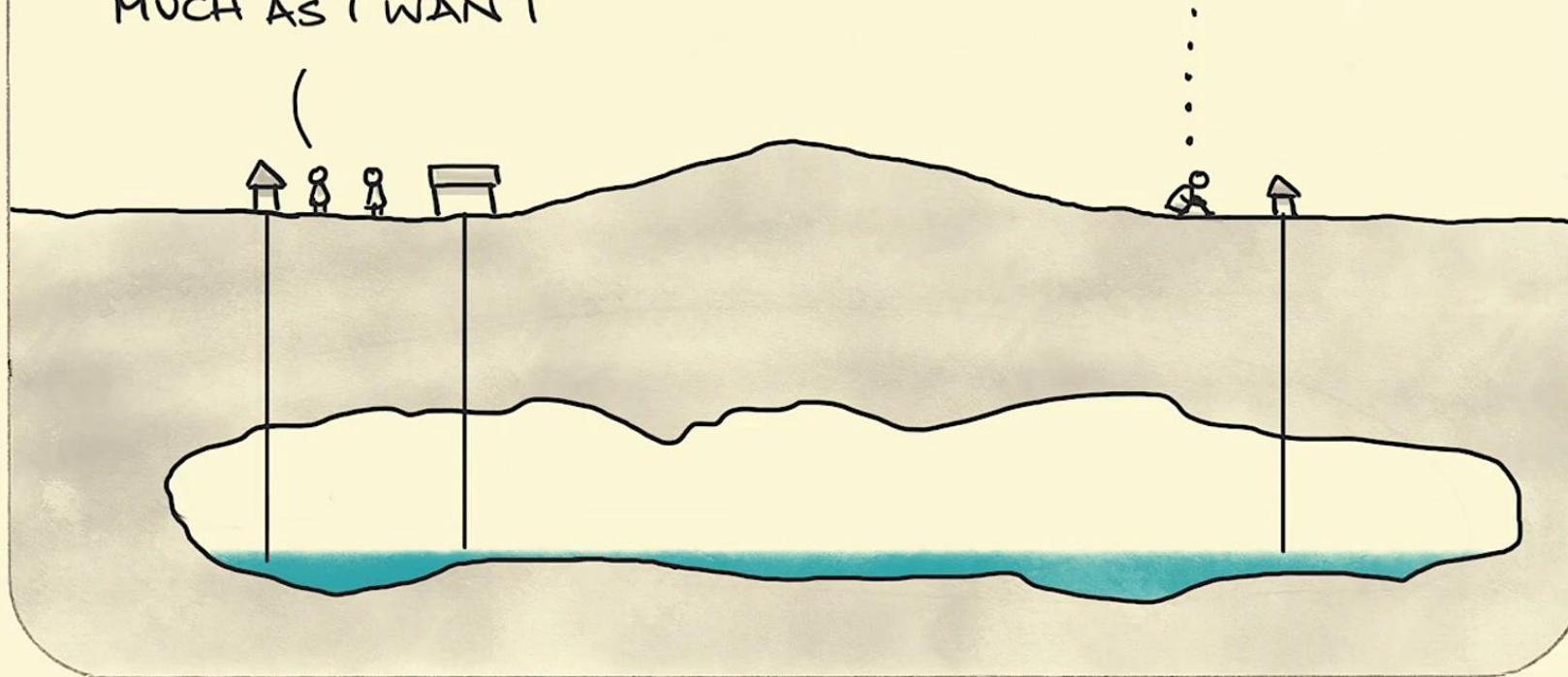
- Or, using algebra, if:
 - $MB_A = 6 - Q$
 - $MB_B = 10 - 2Q$
 - $MC = 2Q$
- Find socially optimal quantity (Q^*)
- **Step 1:**
 - Find “kink” in SMB. This occurs when the “lower” MB curve intersects with the x-axis, or $MB_A = 0$.
 - $0 = 6 - Q \Rightarrow Q = 6$
- **Step 2A:**
 - Find equation for SMB for $Q \geq 6$. This is just MB_B
- **Step 2B:**
 - Find equation for SMB for $Q \leq 6$.
 - Add MB_A and MB_B vertically.
 - $SMB = MB_A + MB_B = (6 - Q) + (10 - 2Q) = 16 - 3Q$
- **Step 3 (find Q^*)**
 - Set $SMB = MC$
 - $16 - 3Q = 2Q \Rightarrow Q^* = 16/5 \approx 3.2$
 - Make sure this is on the right “segment” of SMB!

THE TRAGEDY OF THE COMMONS

WHEN SHORT-TERM SELF-INTEREST LEADS TO TRAGEDY FOR ALL

WATER BELONGS TO
ANYONE. I TAKE AS
MUCH AS I WANT

SEEMS THERE'S LESS
AND LESS THESE DAYS



sketchplanations

- Economic activity treats the atmosphere as a global commons.
- By burning fossil fuels, we're emitting GHGs as if we're overgrazing a common pasture or withdrawing water from a common aquifer.
- Our individual incentives for each GHG emission do not align with what is optimal for global GHG concentrations in the atmosphere...
- ...yet our costs of "over-using" the atmosphere are shared.
- Same problem with free-riding for public goods (in reverse).

For next week...

www.caseyjwichman.com/econ4210#week4